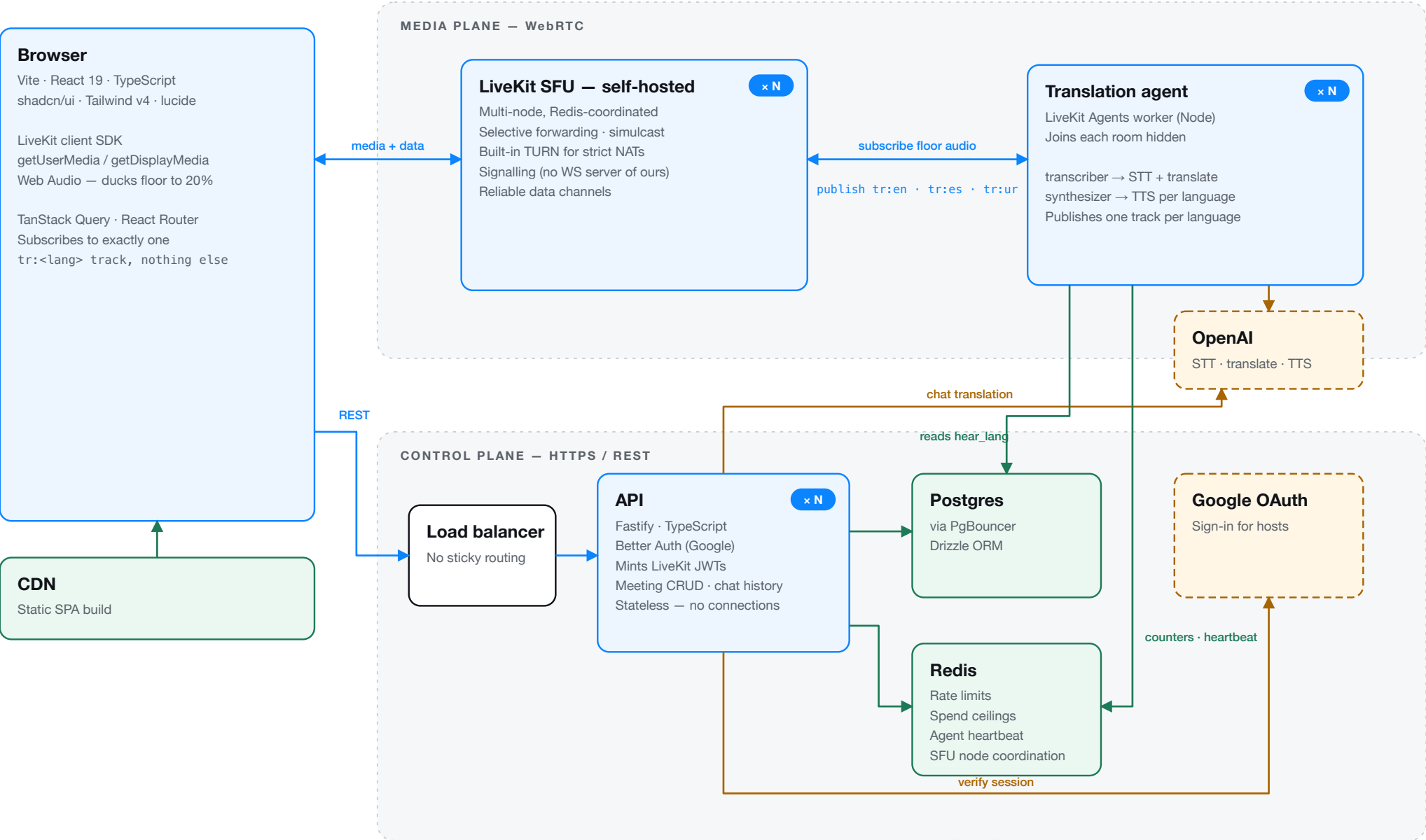


Koine — System Architecture

Real-time meetings with simultaneous translation · spec 2026-09-18 · self-hosted LiveKit · every meeting uses the SFU, at every size

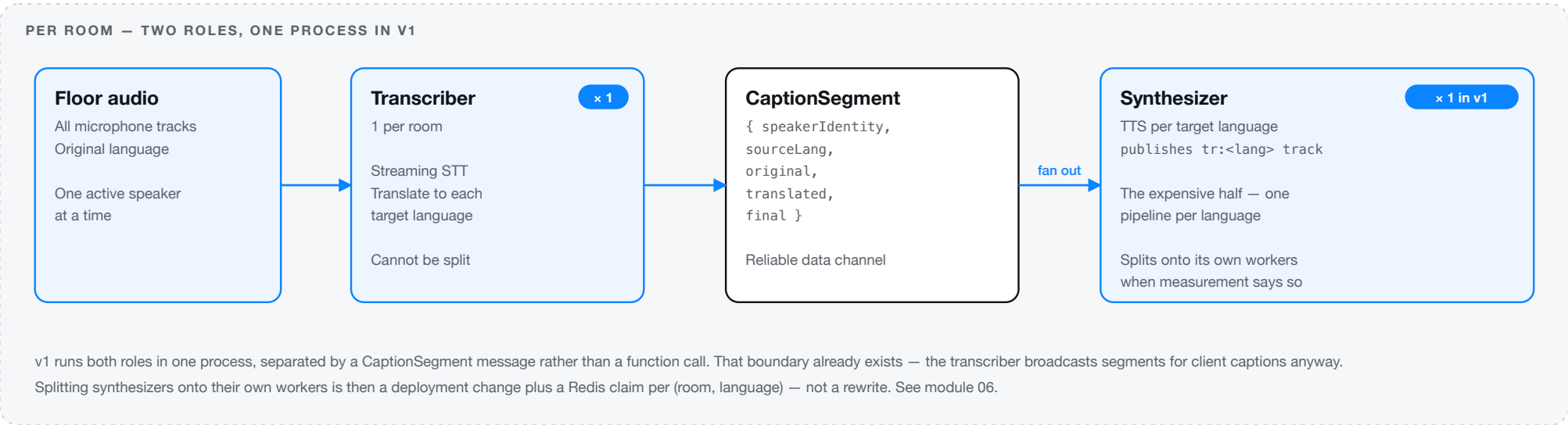


 Ours — scales horizontally State Third party

Limit, in two places: a room lives on one SFU node, and its translation runs on one agent worker. Rooms distribute across nodes and workers; a room does not split across either. Peer-to-peer was rejected — module 05.

Koine — Translation Pipeline

One channel per target language, not per listener · cost scales with language diversity, not headcount



Channel derivation

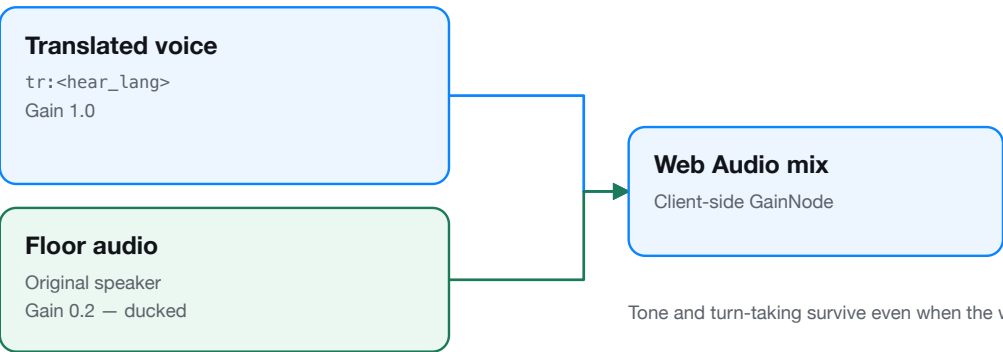
```
channels = distinct(participant.hear_lang)
- {'floor'} - {meeting.floor_lang}
```

Stored in Postgres, so a replacement worker rebuilds it without client help.

What it costs

25 people, 1 language	0 channels
25 people, 4 languages	4 channels
6 people, 6 languages	6 channels
Naive per-listener build	25 channels

At the listener



Latency budget



Target < 1.5 s · hard ceiling 2.5 s
Past the ceiling this stops being a conversation and becomes a briefing.
Highest risk in the project. Measure before building UI on top of it.

If TTS fails, captions continue and floor audio un-ducks.
Translation is an enhancement layer; it never stops the call.